

Angle between 2 lines and Trigonometry of small angle

- Find, to the nearest degree, the acute angle between two lines whose gradients are:
 - $m_1 = 2$ and $m_2 = \frac{1}{2}$
 - $m_1 = 4$ and $m_2 = -2$
 - $m_1 = 1\frac{1}{4}$ and $m_2 = \frac{1}{3}$
- Find the acute angle between each pair of lines:
 - $y = 3x - 1$ and $y = -2x + 5$
 - $x - 2y + 1 = 0$ and $x + 3y + 2 = 0$
 - $4x - 3y + 2 = 0$ and $7x + y - 1 = 0$
 - $4x + 3y + 5 = 0$ and $3x - 4y - 2 = 0$
 - $y - \sqrt{3}x - 5 = 0$ and $\sqrt{3}y - x + 6 = 0$
 - $y + 4x - 7 = 0$ and $y = 5 - 4x$
- Find, to the nearest minute, the acute angle between the lines $3x + 4y = 8$ and $2x + 3y = 5$.
- Use the formula $\tan \theta = \left| \frac{1}{m} \right|$ to find, correct to the nearest minute, the acute angle between the vertical line $x = -1$ and the line $4x - 3y + 5 = 0$.
- Find the point of intersection of the parabolas $y = x^2$ and $y = (x - 2)^2$.
 - Sketch the two curves, showing their point of intersection.
 - Find the acute angle, to the nearest degree, at which the curves intersect.
- Show that the curves $y = e^x$ and $y = x^2 - 3x + 1$ have a common point at $(0, 1)$.
 - Find, to the nearest degree, the acute angle between the tangents to the curves at $(0, 1)$.
- Find the acute angle and the obtuse angle between the lines:
 - $2x - y + 1 = 0$ and $11x - (8 + 5\sqrt{3})y + 5\sqrt{3} = 0$
 - $y = mx + b$ and $(m - 1)x - (m + 1)y = (m^2 - 1)b$
- A, B and C are the point $(-2, 1)$, $(1, 7)$ and $(5, 0)$ respectively. Calculate the three interior angles of $\triangle ABC$ correct to the nearest minute and check their sum. Explain why the sum is not exactly 180°
- The line $y = mx$ is inclined at 45° to $y = 2x - 4$, Find the two possible values of m .
 - Explain geometrically why these two values of m represent perpendicular gradients.
- Show that the line $y - 1 = m(x - 1)$ always passes through the point $(1, 1)$.
 - Find, in general form, the equations of the two lines through $(1, 1)$ that are inclined at 45° to the line $y = 5x + 6$.
- Find, in general form, the equations of the lines that pass through the point $(1, 2)$ and are inclined at an angle of 60° to the line $x + y = 3$.
 - Find, in general form, the equations of the lines that pass through $(2, 3)$ and are inclined at an angle whose tangent ratio is $\frac{1}{3}$ to the line $y = 2x + 4$.
- Sketch the parabolas $y = 3x^2$ and $y = 4x - x^2$, showing their points of intersection.
 - Find, to the nearest minute, the acute angles between these curves at their points of intersection.
- Show that the curves $y = \ln(x - 1)$ and $y = \frac{2-x}{x}$ meet at $(2, 0)$.
 - Find $\tan \alpha$, where α is the acute angle between the tangents at $(2, 0)$.
- Find the points of intersection of $y = x^2$ with the curve defined parametrically by $x = t^3$ and $y = t^2$. Find the angle between the curves at these points. Sketch the situation.
- Find the following limits
 - $\lim_{x \rightarrow 0} \frac{\sin x}{x}$
 - $\lim_{x \rightarrow 0} \frac{\sin 2x}{x}$
 - $\lim_{x \rightarrow 0} \frac{\sin x}{2x}$
 - $\lim_{x \rightarrow 0} \frac{\sin 3x}{2x}$
 - $\lim_{x \rightarrow 0} \frac{5x}{\sin 3x}$
 - $\lim_{x \rightarrow 0} \frac{\sin 3x + \sin 5x}{x}$
- Find the following limits:
 - $\lim_{\theta \rightarrow 0} \frac{\tan \theta}{\theta}$
 - $\lim_{\theta \rightarrow 0} \frac{\tan 7\theta}{5\theta}$
 - $\lim_{\theta \rightarrow 0} \frac{7\theta}{3 \tan 5\theta}$

17. Find

- $\lim_{x \rightarrow 0} \frac{\sin x}{\frac{x}{2}}$
- $\lim_{x \rightarrow 0} \frac{\sin \frac{x}{2}}{x}$
- $\lim_{x \rightarrow 0} \frac{\sin \frac{x}{3}}{2x}$

18. Find

- $\lim_{x \rightarrow 0} \frac{\sin 2x}{\sin x}$
- $\lim_{y \rightarrow 0} \frac{\tan 3y}{\tan 2y}$
- $\lim_{t \rightarrow 0} \frac{\sin 5t}{\tan 7t}$

19. Find

- $\lim_{\theta \rightarrow 0} \frac{1 - \cos^2 \theta}{\theta^2}$
- $\lim_{\theta \rightarrow 0} \frac{\tan a\theta}{\sin b\theta}$
- $\lim_{\theta \rightarrow 0} \frac{\theta^2}{1 - \sin^2 \theta}$

20. Show that

- $\lim_{n \rightarrow 0} \frac{\sin n^\circ}{n} = \frac{\pi}{180}$
- $\lim_{n \rightarrow \infty} n \sin \frac{\pi}{n} = \pi$

21. a. Write down the compound angle formula for $\cos(A + B)$.

b. By replacing both A and B with x , show that $\cos 2x = 1 - 2 \sin^2 x$.

c. Hence find $\lim_{x \rightarrow 0} \frac{1 - \cos 2x}{x^2}$

22. a. Express 2° in radians.

b. Explain why $\sin 2^\circ \approx \frac{\pi}{90}$

c. Taking π as 3.142, find $\sin 2^\circ$ correct to four decimal places *without* using a calculator.

23. A car travels 1 km up a road which is inclined at 5° to the horizontal. Through what vertical distance has the car climbed? (Use the fact that

$\sin x \doteq x$ for small angles, and give your answer correct to the nearest metre.)

24. A tower is 30 metres high. What angle, correct to the nearest minute, does it subtend at a point 4 km away? (Use the fact that when x is small, $\tan x \doteq x$.)

25. The moon subtends an angle of $31'$ at an observation point on Earth 400000 km away. Use the fact that the diameter of the moon is approximately equal to an arc of a circle whose centre is the point of observation to show that the diameter of the moon is approximately 3600 km.

26. A regular polygon of 300 sides is inscribed in a circle of radius 60 cm. Show that each side is approximately 1.26 cm.

27. a. Show that $\lim_{x \rightarrow 0} \frac{1 - \cos x}{x^2} = \frac{1}{2}$ by multiplying top and bottom by $1 + \cos x$.

b. Explain why this means that $\cos x \doteq 1 - \frac{1}{2}x^2$ for small angles.

c. Use the calculator to approximate the difference between $\cos x$ and $1 - \frac{1}{2}x^2$ for small angles, and find the smallest angle, in degrees for which this difference exceeds 0.001.

d. A post 5 metres long leans at 10° to the vertical. Find how high the top is above the ground using the approximation $\cos x \doteq 1 - \frac{1}{2}x^2$. Give your answer correct to four significant figures, and then check using the calculator's cosine function.

Answer

- a. 37° ; b. 41° ; c. 33°
- a. 45° ; b. 45° ; c. 45° ; d. 90° ; e. 30° ; f. The lines are parallel and distinct, and so do not intersect at all.
- $3^\circ 11'$
- $36^\circ 52'$
- a. (1, 1); c. 53°
- b. 63°
- a. $30^\circ, 150^\circ$; b. $45^\circ, 135^\circ$
- $\angle A = 71^\circ 34'$, $\angle B = 56^\circ 19'$ and $\angle C = 52^\circ 8'$. The sum is $180^\circ 1'$. The error is due to rounding.
- a. $m = -3$ or $\frac{1}{3}$; b. Let A and B be the points where $y = -3x$ and $y = \frac{1}{3}x$ meet $y = 2x - 4$. $\triangle AOB$

has two 45° angles, so the third angle, $\angle AOB$, is 90° .

10. b. $3x + 2y - 5 = 0$, $2x - 3y + 1 = 0$

11. a. $(2 - \sqrt{3})x - y + \sqrt{3} = 0$, $(2 + \sqrt{3})x - y - \sqrt{3} = 0$; b. $x - y + 1 = 0$, $7x - y - 11 = 0$

12. b. $75^\circ 58'$ at (0, 0) and $17^\circ 6'$ at (1, 3)

13. b. 3

14. At (0, 0) the curves are perpendicular. At (1, 1) and at (-1, 1), $\tan \theta = \frac{4}{7}$, so the angle is $29^\circ 45'$

15. a. 1; b. 2; c. $\frac{1}{2}$; d. $\frac{3}{2}$; e. $\frac{5}{3}$; f. 8

16. a. 1; b. $\frac{7}{5}$; c. $\frac{7}{15}$

17. a. 2; b. $\frac{1}{2}$; c. $\frac{1}{6}$

18. a. 2; b. $\frac{3}{2}$; c. $\frac{5}{7}$

19. a. 1; b. $\frac{a}{b}$; c. 0

21. a. $\cos A \cos B - \sin A \sin B$; c. 2

22. a. $\frac{\pi}{90}$; b. $\sin 2^\circ = \sin \frac{\pi}{90} \doteq \frac{\pi}{90}$; c. 0.0349

23. 87 metres

24. $26'$

27. c. 23° ; d. 4.924 metres